

UNIT-5

BDA & AI APPLICATIONS

Advanced Topics and Case Studies

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Deep Learning on Big Data using TensorFlow on Spark

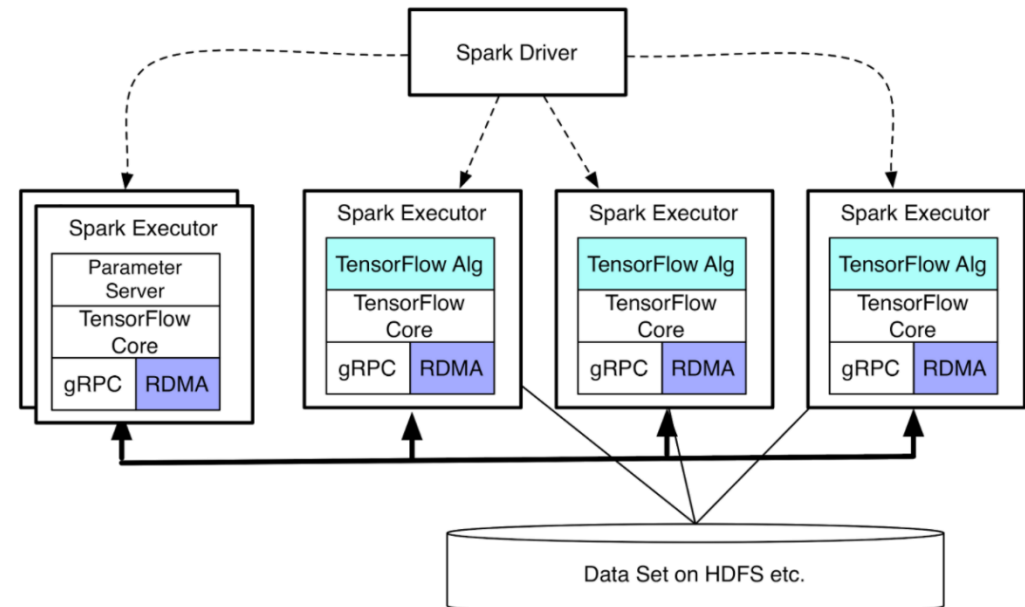
- Deep Learning requires massive datasets to be accurate.
- Standard TensorFlow runs on a single machine (or single GPU cluster).
- **Limitation:** Memory constraints (Out of Memory errors) and impossibly long training times (weeks/months).

The Solution – Best of Both Worlds

- **Apache Spark:** Handles Data Ingestion, Cleaning, and Feature Engineering across a cluster.
- **TensorFlow:** Handles complex tensor mathematics and model optimization.
- **The Goal:** Bring the deep learning code directly to the data, rather than moving the data to the code.

TensorFlowOnSpark (TFoS) Architecture

- Developed and open-sourced by Yahoo.
- Bypasses the need to save Spark data to disk before feeding to TensorFlow.
- Uses Spark's RDDs/DataFrames to stream data directly into TensorFlow memory in real-time.

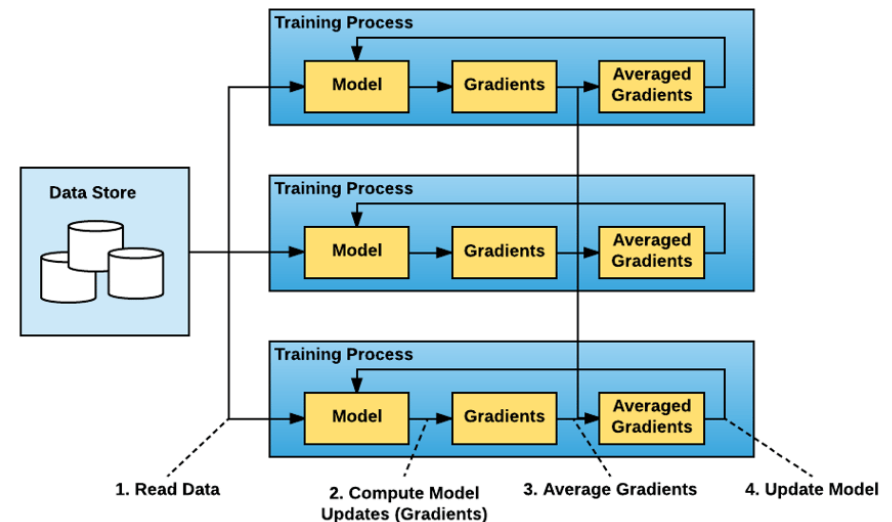


The Distributed Training Workflow

- **Step 1:** Spark loads raw data (HDFS/S3).
- **Step 2:** Spark cleans data, handles missing values, and creates vector formats.
- **Step 3:** Spark launches a TensorFlow cluster alongside its own workers.
- **Step 4:** TF workers compute gradients in parallel and update a central Parameter Server.

Modern Alternatives (Horovod)

- While TFoS is foundational, Horovod (by Uber) is the modern industry standard.
- Uses Ring-AllReduce architecture instead of a central parameter server.
- Requires minimal code changes to scale a standard TF script to hundreds of GPUs.



Explainable AI (XAI) in Big Data Environments

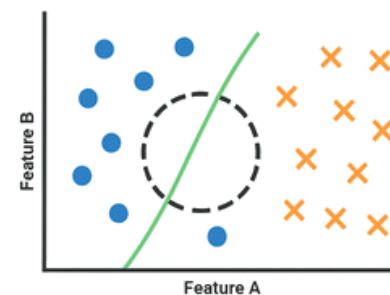
- The "Black Box" Dilemma
- **The Tradeoff:** As AI models become more complex and accurate, they become less interpretable.
- **The Risk:** Hidden biases, spurious correlations, and catastrophic edge-case failures.
- **The Barrier:** Lack of user trust prevents AI adoption in high-stakes fields.

What is Explainable AI (XAI)?

- **Global Explanations:** Understanding the overall behavior and rules of the entire model.
- **Local Explanations:** Understanding why the model made one specific decision for one specific data point.
- **Goal:** Ensure AI is fair, unbiased, and aligned with human values.

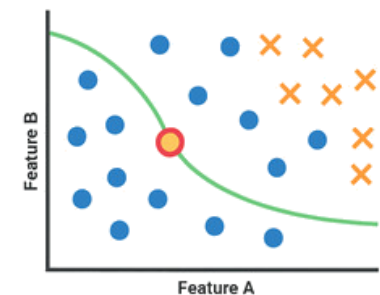
Key XAI Techniques: SHAP and LIME

- **SHAP (SHapley Additive exPlanations)**: Based on game theory; calculates the exact mathematical contribution of every feature to the final prediction.
- **LIME (Local Interpretable Model-agnostic Explanations)**: Tweaks the input data slightly to see how the prediction changes, creating a simple map of the decision.



LIME

- Local explanations
- Linear approximation
- Instability



SHAP

- Global explanations
- Shapley values
- Consistency

The Big Data Challenge for XAI

- XAI algorithms are incredibly computationally expensive.
- Calculating exact SHAP values for massive datasets is an NP-hard problem.
- Running XAI sequentially on a single node is impossible for Big Data.

Scaling XAI with Apache Spark

- **Distributed Explainability:** Using Spark UDFs (User Defined Functions) to run SHAP/LIME across a cluster.
- **Sampling Strategies:** Explaining a statistically significant sample of the Big Data instead of every single row.
- **Libraries:** Integrating tools like shap with PySpark DataFrames.

Ethical Issues and Data Governance in Big Data AI

- **The Core Ethical Dilemmas in Big Data:** Privacy, Bias, and Consent.
- **Privacy vs. Utility:** The tension between extracting valuable insights and protecting individual anonymity.
- **Algorithmic Bias:** Machine Learning amplifies historical human prejudices.
- **Informed Consent:** Do users actually understand what they are agreeing to in 50-page Terms of Service agreements?

Case Studies in Algorithmic Bias

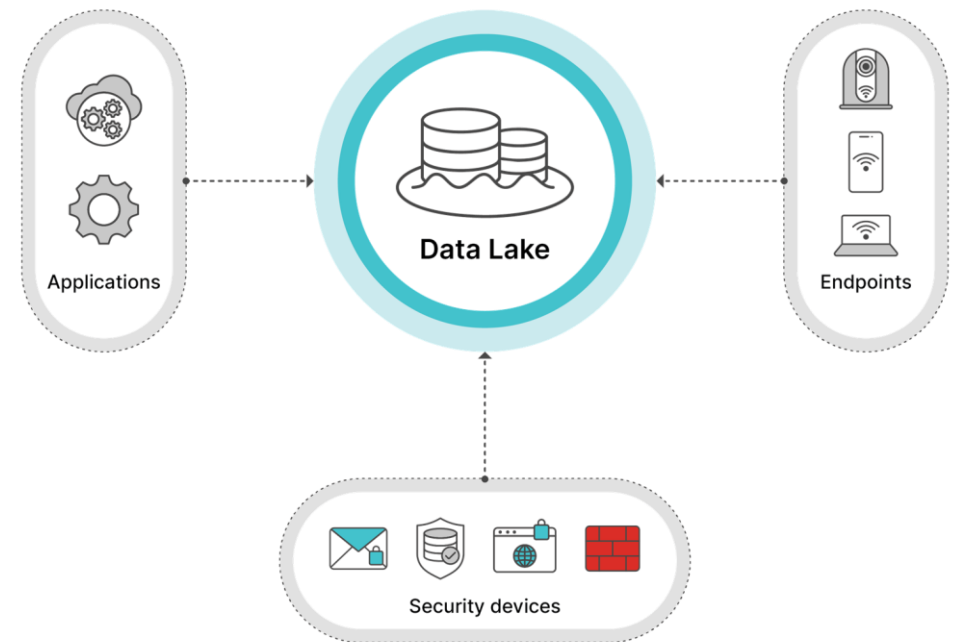
- **The Recruiting AI:** Trained on 10 years of historical resumes (mostly male); the AI learned to penalize female candidates.
- **Predictive Policing:** Historical arrest data leads algorithms to over-police minority neighborhoods, creating a self-fulfilling feedback loop.
- **The Lesson:** "Garbage In, Garbage Out" applies to ethics, not just data quality.

The Myth of Anonymization

- Removing Personally Identifiable Information (PII) like Names and SSNs is not enough.
- **The Netflix Prize Leak:** Anonymous movie ratings were cross-referenced with public IMDb reviews to identify specific users and their private viewing habits.
- **The Solution:** Advanced techniques like Differential Privacy and k-anonymity.

What is Data Governance?

- It is not just IT; it is a business strategy.
- **Data Stewardship:** Assigning clear owners who are responsible for specific datasets.
- **Data Lineage:** Tracking data from its origin, through all transformations, to its final AI model.
- **Master Data Management (MDM):** Ensuring a single, accurate version of the truth across the enterprise.



Regulatory Frameworks (GDPR, CCPA, DPDP)

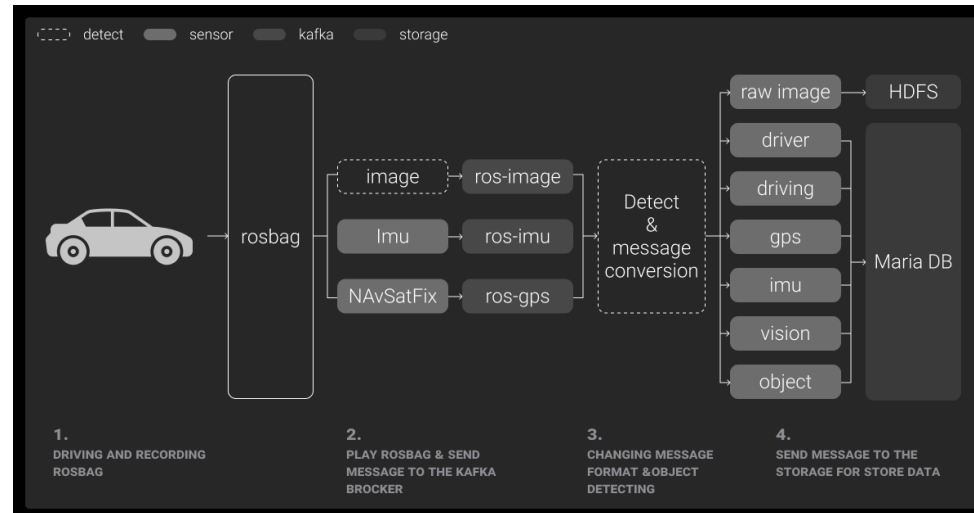
- **GDPR (Europe):** The gold standard. Introduces massive fines, the "Right to be Forgotten," and strict consent rules.
- **CCPA (California):** Focuses heavily on the right to opt-out of data sales.
- **DPDP Act (India):** India's Digital Personal Data Protection Act, focusing on data minimization and purpose limitation.

Technical Implementation in Big Data

- **Apache Ranger:** Centralized security framework to manage access control (who can read/write what) across the Hadoop ecosystem.
- **Apache Atlas:** Data governance and metadata framework to track data lineage and catalog assets.
- **Audit Logs:** Immutable records of exactly which user or AI algorithm touched a piece of sensitive data.

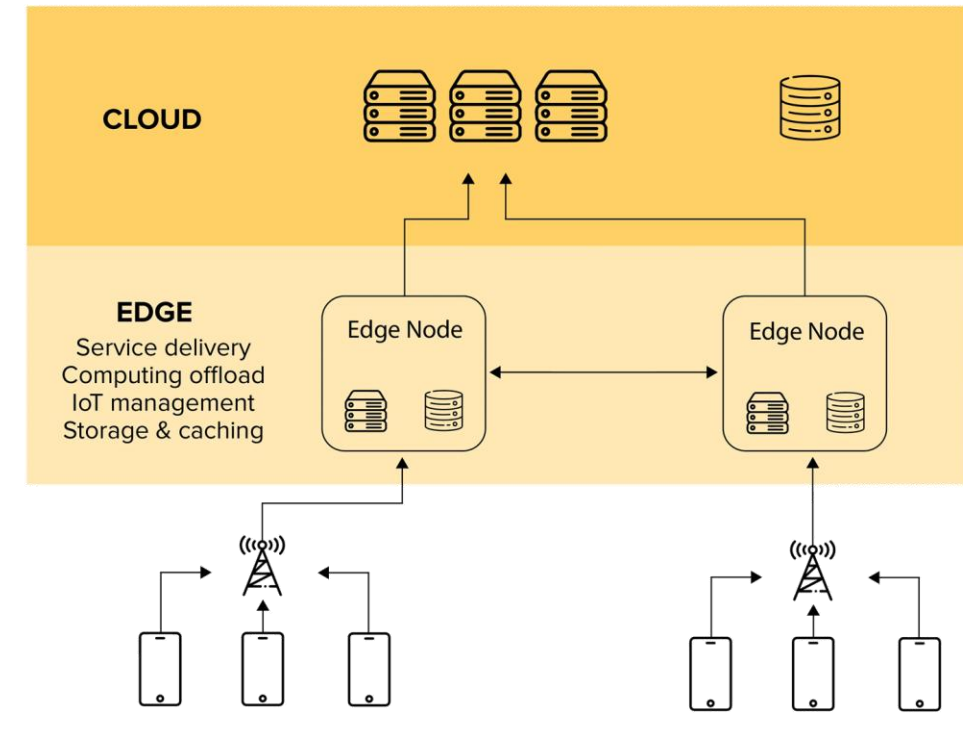
Edge Computing and AI for Low Latency Applications

- The "Round-Trip" Latency Problem
- **Network Latency:** The physical time it takes data to travel.
- **Bandwidth Choke:** Transmitting Terabytes of raw 4K video/sensor data is impossible over mobile networks.
- **Connectivity:** What happens if the network goes down?



What is Edge AI?

- **Training vs. Inference:** We do not train models on the Edge.
- **The Workflow:** Train massive models on Cloud Big Data -> Freeze the model -> Deploy to the Edge.
- **Goal:** Sub-millisecond decision making entirely on local hardware.



How do we fit Big Data AI on tiny devices?

- Edge devices (cameras, sensors) have low RAM, weak CPUs, and run on batteries.
- **Model Quantization:** Reducing the precision of the math (e.g., changing 32-bit decimals to 8-bit integers) to save memory.
- **Model Pruning:** Deleting artificial neurons that don't contribute much to the final answer.

Specialized Edge Hardware

- Standard CPUs are not designed for AI matrix math.
- **NPU (Neural Processing Units)**: Chips specifically designed to run AI inference at ultra-low power.
- **The "System on a Chip" (SoC)**: Bringing the camera, memory, and AI brain onto a single board.

The Edge-to-Cloud Big Data Pipeline

- The Edge does not replace the Cloud; it acts as a smart filter.
- Edge AI processes 99% of "normal" data locally and deletes it.
- It only sends anomalies, metadata, or extremely valuable data back to the Big Data Cloud for long-term storage and retraining.

Case Study 1: AI-Powered Big Data in Healthcare

- The Healthcare Data Ecosystem (The 4 V's in Action)
- **Volume:** Petabytes of historical patient records.
- **Velocity:** Real-time streaming vitals from ICU monitors and wearables.
- **Variety:** Unstructured text (clinical notes), 3D images (MRI/CT), and time-series data.
- **Veracity:** The critical need for accurate, clean, and life-saving data.

Deep Dive 1 – Predictive Analytics in the ICU

- **The Problem:** Conditions like Sepsis kill rapidly if not caught early.
- **The Big Data Solution:** Streaming real-time ICU vitals through an Apache Kafka/Spark pipeline.
- **The AI Model:** Recurrent Neural Networks (RNNs) analyzing time-series data to spot invisible physiological trends.

Deep Dive 2 – Medical Imaging & Diagnostics

- **The Bottleneck:** Radiologists suffer from high burnout and visual fatigue.
- **The Technology:** Distributed Deep Learning (CNNs/Vision Transformers) trained on millions of historical scans.
- **The Workflow:** AI acts as a "Tireless Triage," instantly flagging the highest-risk scans for human review.

Genomics and Precision Medicine

- **The Scale:** Sequencing a single human genome generates ~100GB of raw data.
- **Population Mining:** Finding hidden correlations between specific genetic mutations and disease risks across millions of patients.
- **The Outcome:** Moving from "One Size Fits All" treatments to personalized, genetically targeted therapies.

Roadblocks: Data Silos and Interoperability

- **Data Silos:** Hospitals use different, proprietary software systems that cannot talk to each other.
- **Interoperability:** The struggle to standardize medical data formats (e.g., FHIR standards).
- **The Impact:** Fragmented patient histories lead to less accurate AI models.

Governance and Ethics in Medicine

- **Privacy:** Strict adherence to HIPAA/GDPR; medical data is the most sensitive data on earth.
- **Bias:** Ensuring AI models are trained on diverse populations, not just wealthy urban demographics.
- **Explainability:** Doctors must be able to trust and verify why the AI recommended a specific surgery (Ties into XAI).

Case Study 2: Big Data AI Solution in Smart Manufacturing

- **Data Sources:** PLCs (Programmable Logic Controllers), vibration sensors, acoustic sensors, and thermal cameras.
- **The Protocol Challenge:** Converting diverse industrial protocols (Modbus, OPC-UA) into Big Data formats (JSON/Avro).
- **Ingestion:** Using Apache Kafka to handle high-frequency sensor streams (thousands of events per second).

Deep Dive 1 – Predictive Maintenance (PdM)

- **The Logic:** Analyzing time-series data to identify "Signatures of Failure."
- **Algorithm:** Using LSTMs (Long Short-Term Memory) or GRUs to find patterns in vibration and temperature.
- **Impact:** Shifting from "Fixing when broken" to "Fixing just before breaking," saving millions in lost production time.

Deep Dive 2 – Computer Vision for Quality 4.0

- **The Challenge:** Human inspectors can only see 80-90% of defects and suffer from fatigue.
- **The Technology:** Deep Learning (CNNs) deployed at the Edge for sub-millisecond inspection.
- **The Workflow:** Every single part is photographed; the AI compares it against a "Perfect Part" model in real-time.

The "Digital Twin" Concept

- A virtual representation of a physical asset updated with real-time Big Data.
- **Simulation:** Testing "What If" scenarios in the digital world before changing the physical machine.
- **Integration:** Syncing Spark streaming data with 3D CAD models.

Edge-to-Cloud Architecture for Industry

- **Edge:** Critical for safety (E-stops, high-speed sorting).
- **Cloud:** Critical for training models across multiple global factories.
- **Latency Requirements:** Decisions must be made in $<10\text{ms}$ for high-speed robotics.

Economic and Operational Impact

- **Energy Optimization:** AI identifies power-hungry machines and optimizes their duty cycles.
- **Supply Chain Integration:** The factory automatically orders its own spare parts when the AI predicts a failure.
- **The Workforce:** Shifting human roles from "Manual Labor" to "AI Oversight and Maintenance."



Thank You